

The University of Nottingham

SCHOOL OF COMPUTER SCIENCE

A LEVEL ONE MODULE, AUTUMN/SPRING SEMESTER 2021-2022

Mathematics for Computer Scientists 1

Time allowed: ONE Hour and THIRTY Minutes

This is an open-book shorter-timed examination. Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced

The full marks are 100. You MUST answer question 1 worthy of 25 marks, and answer only TWO questions from question 2 to question 4, each worthy of 33 marks. 9 marks are allocated for submission in $\LaTeX$

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

K: Knowledge, C: Comprehension, A: Application

Question 1: Computational logic and sets.**[overall 25 marks]**

- a. Answer true or false for the following questions. p and q are Boolean variables. $\mathbf{S}$ is a set. $\sim \mathbf{S}$ is the complement of $\mathbf{S}$. $\emptyset$ is the empty set.

- i). $(p \Rightarrow q) \wedge (p \Leftarrow q) \equiv (p \equiv q)$.
- ii). $\mathbf{S} = \emptyset, |2^{\mathbf{S}}| = 1$.
- iii). $\sim \mathbf{S} \cap \mathbf{S} = \emptyset$.
- iv). $\neg(p \vee q) \equiv \neg p \vee \neg q$.
- v). $\neg(p \Rightarrow q) \equiv p \Leftarrow q$.

[10 Marks]**Answers: (K,A)**

- i). true.
- ii). true.
- iii). true.
- iv). false.
- v). false.

[10 Marks in total, 2 marks each]

- b. Consider any set $\mathbf{A}$ written out as a list of ten elements, of which two of them are exact copies. For example, $\mathbf{A} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 9\}$. First, mathematically prove for any set $\mathbf{A}$ that the order in which the elements are listed out does not matter. Second, mathematically prove for any set $\mathbf{A}$, the individual elements are counted once so that $|\mathbf{A}| = 9$. Full marks using only basic mathematics and notation as given out in the lecture materials (e.g. set comprehension, propositional laws).

[15 Marks]**Answers: (K,C)**

Let the set $\mathbf{A} = \{a_1, a_2, a_3, \dots, a_{10}\}$. Through set enumeration one has

$$\begin{aligned} \mathbf{A} &= \{a_1, a_2, a_3, \dots, a_{10}\} \\ &= \{x \mid x = a_1 \vee x = a_2 \vee x = a_3 \vee \dots \vee x = a_{10} : x\}. \end{aligned}$$

[3 Marks]

Let $i, j \in \mathbb{Z}_1^{10}$. Let $K = (k_i)_{i=1}^{10}$ be the 10-tuple of any possible permutation of $\{1, 2, 3, \dots, 10\}$. Then, to prove that the order in which the elements in $\mathbf{A}$ are listed out does not matter,

$$\begin{aligned} &\{x \mid x = a_1 \vee x = a_2 \vee x = a_3 \vee \dots : x\} \\ &= \{\text{Symmetry of disjunction } p \vee q \equiv q \vee p, \text{ resulting in any permutation} \\ &\quad \text{of the list of disjunctions}\} \\ &\{x \mid x = a_{k_1} \vee x = a_{k_2} \vee x = a_{k_3} \vee \dots : x\}. \end{aligned}$$

[6 Marks]

To prove the individual elements are counted once in **A**,

$$\begin{aligned}
 & \{x \mid x = a_1 \vee x = a_2 \vee x = a_3 \vee \dots : x\} \\
 &= \{\text{Symmetry of disjunction } p \vee q \equiv q \vee p, \text{ such that } a_{k_9} = a_{k_{10}}\} \\
 & \{x \mid x = a_{k_1} \vee x = a_{k_2} \vee x = a_{k_3} \vee \dots \vee x = a_{k_9} \vee x = a_{k_{10}} : x\} \\
 &= \{\text{Idempotence of disjunction } x = a_{k_9} \vee x = a_{k_{10}} \equiv x = a_{k_9}\} \\
 & \{x \mid x = a_{k_1} \vee x = a_{k_2} \vee x = a_{k_3} \vee \dots \vee x = a_{k_9} : x\}.
 \end{aligned}$$

After which, $|\mathbf{A}| = \{\sum x \mid x = a_{k_1} \vee x = a_{k_2} \vee x = a_{k_3} \vee \dots \vee x = a_{k_9} : 1\} = 9$.

[6 Marks]

Note: partial marks for non-formal and basic proofs. This include specific examples or less formal (use of natural language argumentations) proofs are given (although still correct by identifying the relevant mathematical laws).

Question 2: Questions on Functions, Relations, and Matrices.

[overall 33 marks]

a. For each of the following statements, say whether it is **true** or **false**.

- There are in total $|A| \times |B|$ different relations from a finite set A to a finite set B .
- There are in total $|A|^{|B|}$ different total functions from a finite set A to a finite set B .
- Let $\mathcal{Y}$ be the set of real functions. Define the relation R from $\mathbb{R}$ to $\mathcal{Y}$ such that aRf if and only if $f(a) = 0$. Therefore, $-1R(x^2 - 1)$.
- The "is a superset of" relation $\supseteq$ is the inverse of $\subseteq$.
- Define f from $\mathbb{Z}^+$ to $\mathbb{Z}^+$, where $\mathbb{Z}^+$ denotes the set of positive integers, such that $f(n) = 3n^2 - 2$. Based on the definition, f is an injective function.

[5 Marks]

Answers: (K,C)

- False
- False
- True
- True
- False

1 mark each.

b. Let f , g , and h be functions from $\mathbb{R}$ to $\mathbb{R}$, where $\mathbb{R}$ denotes the set of real numbers. Note the definitions of the operations $+$ and $\cdot$ are given in the lectures.

- Show that $(f + g) \circ h = f \circ h + g \circ h$.

[4 Marks]

- ii). The sum of two subsets A and B of $\mathbb{R}$ is defined by $A + B = \{c : \exists a \in A, b \in B, c = a + b\}$. Show that the range($f + g$) $\subseteq$ range(f) + range(g), where range(f) and range(g) denote the range of function f and g , respectively.

[7 Marks]

- iii). Define functions f, g, h by $f(x) = -5x + 2$, $g(x) = 5x^2 - 7$, and $h(x) = (x - 3)/(x + 2)$, respectively. Determine whether each of them is a bijection. Explain your answer.

[3 Marks]

Answers: (C, A)

- i). Show that $\forall x \in \mathbb{R}, (f+g) \circ h(x) = (f+g)(h(x)) = (f(h(x)) + g(h(x))) = f \circ h(x) + g \circ h(x)$. 1 mark for each step, and 1 mark for using the proof technique of universal instantiation.

- ii). $\forall y \in \text{range}(f+g), \exists x \in \mathbb{R}, y = (f+g)(x) = f(x) + g(x)$ (1 mark for universal instantiation, 1 mark for applying $f+g$). Let $y_f = f(x), y_g = g(x)$, therefore $y_f \in \text{range}(f), y_g \in \text{range}(g)$ (2 marks). According to the definition of $A+B$ given above, we have $y_f + y_g \in \text{range}(f) + \text{range}(g)$ (1 mark) and therefore $y \in \text{range}(f) + \text{range}(g)$ (1 mark). According to the definition of subset, we then have $\text{range}(f+g) \subseteq \text{range}(f) + \text{range}(g)$ (1 mark).

- iii). $f(x)$ is one-to-one and onto, so is bijective (1 mark). $g(x)$ is neither one-to-one nor onto and therefore is not bijective (1 mark). $h(x)$ is not onto since it is not defined when $x = -2$ and is therefore not bijective (1 mark).

- c. SoftJobs is a company capable of producing a variety of software packages, ranging from data management software to video games. The company uses a database to keep track of which programmers are working on which projects for which clients, shown in Table 1.

Project	Client
Football Dunkfest	GameCo
Rx Tracker	MediComp
Skate Rats	GameCo

(a) Contracts

Programmer	Project
Nurul	Football Dunkfest
Ali	Rx Tracker
Chan	Football Dunkfest
Nanda	Skate Rats
Rajeswari	Skate Rats

(b) Work Assignments

Table 1: Database for SoftJobs

- i). Represent the relations depicted in Table 1a and 1b with binary matrices, respectively.

[8 Marks]

- ii). Compose the two matrices you obtain in item c.i) and determine for which client Nurul is working based on the result of composition.

[6 Marks]

Answers: (A)

- i). There are 3 projects, 2 clients, and 5 programmers. Let A, B, C be the set of projects, clients, and programmers respectively. The contract relation maps A to B and the binary matrix would consist of 3 rows and 2 columns such that

$$M_{Contract} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

where the rows are Football, Rx, and Skate respectively and the columns are GameCo and MediComp. [4 marks]

The work assignment relation maps C to A and the binary matrix would consist of 5 rows and 3 columns such that

$$M_{WorkAssignment} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

where the rows are Nurul, Ali, Chan, Nanda, and Rajeswar, respectively, and the columns are Football, Rx, Skate, respectively. [4 marks]

- ii). To determine which programmer works for which client, it is to determine the relation from C to B (1 mark). To obtain this relation, we can compose relation Contract with relation WorkAssignment, i.e., $R_{Contract} \circ R_{WorkAssignment}$ (1 mark). The composition can be calculated with the two binary matrices obtained in item i) as follows.

$$\begin{aligned} & R_{Contract} \circ R_{WorkAssignment} \\ &= M_{WorkAssignment} \cdot M_{Contract} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 0 \\ 1 & 0 \end{bmatrix} \end{aligned}$$

The first row of the resultant matrix represents Nurul is working for GameCo. [3 marks to equations and 1 mark for the final declaration]

Question 3: Questions on Numbers and Matrices.

[overall 33 marks]

- a. For each of the following statements, say whether it is **true** or **false**.

- i. Let a and m be integers with $m \geq 1$, then $a|a^m$.
- ii. 2 is the only even prime number.

- iii. $\forall x, y \in \mathbb{R}, \lfloor x + y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor$.
- iv. 35 hours after 2pm would be 1pm.
- v. $\gcd(111, 51) \equiv 3 \pmod{3}$

[5 Marks]

Answers: (K, C)

- i). True
- ii). True
- iii). False
- iv). False
- v). True

- b. Show that, if an integer $n > 1$ is composite, then n has a factor larger than 1 and less than or equal to $\sqrt{n}$.

[5 Marks]

Answers: (C)

The following is an example based on proof-by-contradiction.

Since $n > 1$ is composite, we can always factorise n as the product of two positive factors $n = a \cdot b$ where $a \geq 1, b \geq 1$. (1 mark) Without loss of generality, we can assume that $a \leq b$. Assume that there is no factor that is larger than 1 and less than or equal to $\sqrt{n}$ (1 mark). This leads to that $a > \sqrt{n} \wedge b > \sqrt{n}$ (1 mark), which means that $n = ab > \sqrt{n} \cdot \sqrt{n} = n$, i.e., $n > n$ (1 mark). By contradiction, it means our assumption that there is no factor that is larger than 1 and less than or equal to $\sqrt{n}$ is wrong, hence the proof (1 mark).

Other valid proofs are also acceptable.

- c. Suppose n beads are to be arranged on a circular loop, and every k th bead must be the same colour. That is, for each bead b , both beads k positions away from b must be the same colour as b . Figure 1 shows a case where $n = 10$ and $k = 4$. In general, for a cycle of n beads and a skipping number k , what is the maximum number of different colours that could be used? Explain. (Hint: $nk = \text{lcm}(n, k) \cdot \gcd(n, k)$)

[10 Marks]

Answers: (C,A)

Since the beads are arranged in a circle, whichever bead to start with the colouring does not matter (1 mark).

For a bead in a start position, to find all the beads that should be coloured the same as the starting bead, we need to check every k positions either clockwise or counter-clockwise until we come back to the starting bead. (2 mark)

This process is equivalent to finding the least common multiple $\text{lcm}(n, k)$ of n and k . (2 marks)

This means the total number of beads that would have the same colour as the starting

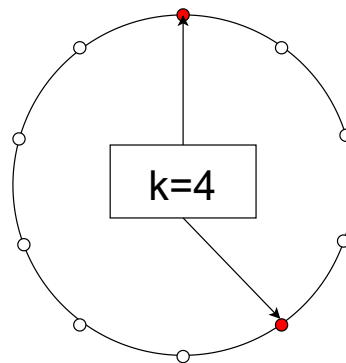


Figure 1: Cycle of $n = 10$ beads with skipping number $k = 4$

bead is $a = \frac{\text{lcm}(n,k)}{k}$ (2 mark)

Recognising that if we choose the next immediate neighbor of the starting bead, we can obtain another set of beads of number a that would have a different colour from that of the first set. This would go on until we hit the bead that share the same colour of the bead at the starting position of the first set. We can obtain in total $b = \frac{n}{a} = \frac{n \cdot k}{\text{lcm}(n,k)}$ (2 marks)

Note that $nk = \text{lcm}(n,k) \cdot \text{gcd}(n,k)$, therefore we have $b = \text{gcd}(n,k)$ (1 mark)

Other alternative solutions are also acceptable.

For example, assuming the total number of different colours is c , one may directly recognise that c must divide n , i.e., c is a divisor of n . It is also the case that c is a divisor of k , and c is a divisor of the remainder of $r = n \bmod k$. This shows that c must be a common divisor of n, k, r , which further implies that c is the greatest common divisor of n, k , i.e., $c = \text{gcd}(n, k)$

- d. Given a permutation $p_1, p_2, \dots, p_n$ of the integers $1, 2, \dots, n$, the corresponding permutation matrix is the n -by- n matrix $P_n = [a_{i,j}]$ such that

$$a_{i,j} = \begin{cases} 1 & \text{if } p_i = j \\ 0 & \text{otherwise} \end{cases}$$

- i. Determine the permutation matrix according to the permutation 3, 2, 5, 4, 1.

[4 Marks]

- ii. Show that for any n , $P_n \cdot P_n^T = I_n$ where I_n is an n -by- n identity matrix.

[9 Marks]

Answers: (K,C)

- i). The permutation matrix is as follows

$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

4 marks.

- ii). Firstly, determine the transpose of the matrix P_n as $P_n^T = [b_{i,j}]$ such that $b_{i,j} = a_{j,i}$ where $a_{j,i}$ is defined in the equation above. (2 marks)

According to the definition of matrix multiplication and assuming the resultant matrix of $P_n \cdot P_n^T$ is A , so the i -th row and j -th column of A is calculated as $c_{i,j} = \sum_{k=1}^n (a_{i,k} \cdot b_{k,j})$ (2 marks)

Note that $\forall i \in \{1, 2, \dots, n\}$ there is only one $a_{i,k_i} = 1$ and $\forall i \neq j, k_i \neq k_j$ (2 marks)
Therefore

$$a_{i,k} \cdot b_{k,j} = \begin{cases} 1 & \text{if } k = k_i \wedge i = j \\ 0 & \text{otherwise} \end{cases}$$

(2 marks)

This leads to $c_{i,i} = 1$ and $\forall i \neq j, c_{i,j} = 0$, hence $C = I_n$ (1 mark).

Question 4: The following questions are pertaining to Counting and Probability.

[overall 33 marks]

a. For each of the following statements, say whether it is **true** or **false**.

- i. There are in total $10 \cdot P(9, 4)$ possible different number plates consisting of 6 digits (0 to 9) with exactly 2 of the same digit.
- ii. Three standard (6-sided) dice are rolled. One is red, one is blue, and one is green. If the particular number showing on each coloured die is important, then there are in total 6^3 possible different outcomes.
- iii. A bag contains 10 black and 12 white balls and each ball is considered distinguishable from each other. There are in total $C(10, 5)C(12, 3) \cdot C(10, 6)C(12, 2)$ different ways to select 8 balls so that exactly 5 or 6 are black.
- iv. Two 6-sided fair dice are rolled. The probability that at least one of the dice is 6, is $\frac{1}{3}$.
- v. Tossing a fair coin and getting a head is mutually exclusive from rolling a 6-sided die and getting a 5.

[10 Marks]

Answers: (K,C)

- i). False
- ii). True
- iii). False
- iv). False
- v). False

(2 marks each)

- b. A necklace is to be formed by threading 8 beads on a circle of string. The beads are taken from a bag containing 14 different beads. How many options are there for a necklace if the knot holding the string together is not visible? Explain your answer.

[5 Marks]

Answers: (C,A)

Choosing 8 beads out of 14 beads and putting them in sequence is a problem of choosing r out of n permutation, the formula of which is $P(n,r)$. In this particular case, $n = 14, r = 8$, therefore we have in total $P(14,8)$ ways in choosing 8 beads out of 14 beads and putting them in a sequence.

On the other hand, since the sequence is now putting in a ring/circle, any of the 8 bead could be the start of the ring but they denote the same ring. Therefore the results obtained for forming the sequence should be divided by 8, hence the final result $\frac{P(14,8)}{8}$.

3 marks for calculation and 2 marks for explanation.

- c. An insurance company issues car insurance, homeowner insurance, and life insurance. Of the claims the company pays, 50% are car, 40% are homeowner, and 10% are life. The company is primarily concerned with claims over RM50,000 that it pays. History has shown that 10% of car claims, 30% of homeowner claims, and 90% of life insurance claims exceed RM50,000.

- i. What percentage of claims paid by this company exceed RM50,000?

[12 Marks]

- ii. Given that this company pays a claim over RM50,000, what is the probability that it is a homeowner claim?

[6 Marks]

Answers: (A)

- i). Firstly, to formalise the problem let us denote the random variable of insurance claim as X , and the sample space of X is $S = \{car, homeowner, life\}$. (1 mark)

Therefore the marginal probabilities are $P(X = car) = 0.5, P(X = homeowner) = 0.4, P(X = life) = 0.1$ (3 marks)

Denote the random variable of the amount of the insurance claim as Y and the sample space of Y is $T = \{exceedRM50,000, notexceedRM50,000\}$ (1 mark)

According to the given conditions we have $P(Y = exceedRM50,000|X = car) = 0.1, P(Y = exceedRM50,000|X = homeowner) = 0.3, P(Y = exceedRM50,000|X = life) = 0.9$ (3 marks)

Therefore that the probability that the claims paid by the company that exceeds RM50,000 is

$$\begin{aligned}
 & P(Y = exceedRM50,000) \\
 &= \sum_{X \in S} P(Y = exceedRM50,000, X) \\
 &= \sum_{X \in S} P(Y = exceedRM50,000|X) \cdot P(X) \\
 &= P(Y = exceedRM50,000|X = car) \cdot P(X = car) \\
 &\quad + P(Y = exceedRM50,000|X = homeowner) \cdot P(X = homeowner) \\
 &\quad + P(Y = exceedRM50,000|X = life) \cdot P(X = life) \\
 &= 0.1 \times 0.5 + 0.3 \times 0.4 + 0.9 \times 0.1 \\
 &= 0.26
 \end{aligned}$$

(4 marks)

ii). Applying Bayes Theorem, we have

$$\begin{aligned} & P(X = \text{homeowner} | Y = \text{exceedRM50,000}) \\ &= \frac{P(Y = \text{exceedRM50,000} | X = \text{homeowner}) \cdot P(X = \text{homeowner})}{P(Y = \text{exceedRM50,000})} \\ &= \frac{0.3 \times 0.4}{0.26} \\ &= \frac{6}{13} \end{aligned}$$

(2 marks for applying Bayes theorem, 4 marks for calculation)